

Ying Xiang Seah

Full-Stack Developer

Computer Science and Design Undergraduate @ Singapore University of Technology and Design



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Work Experience

Mixed Reality Developer • HelloHolo • Singapore

Full-time • Feb 2024 - Aug 2024 • 6 mo

- Spearheaded development of an internal UI Toolkit in Unity to streamline XR workflows, replacing the old and unsupported Mixed Reality Toolkit (MRTK).
 - Advanced **Unity** editor scripting.
 - Created comprehensive API documentation using DocFX.
- Served as **trainer** for DSTA BrainHack XRperience 2024 for over 120 participants
 - Created and delivering **Unity** and **C#** instructional materials for **cross-platform XR** content creation.
 - Demonstrated Apple Vision Pro's capabilities and provided development support.

Part-time • Oct 2020 - Oct 2022 • 2 yr

- Assistant trainer** for immersive media courses with over 150 students for several major local education institutions.
- Built a **digital twinning** experience for National University of Singapore's 5G Lab exhibition.
- Worked with Microsoft Egypt to deliver an interactive data visualization app that displays **dynamic real-time IOT data** on construction analytics for a prominent national project.
- Delivered multiple MR-based **training simulators** on HoloLens 2 for various prominent global corporations for their internal employee training initiatives.
- Assisted as a **technical consultant** for a student research project from a renowned local technological university, exploring the use of **digital twinning in MR** to aid engineers in testing and evaluating mechanical designs and assemblies from CAD software.

Projects

F1gures - Home Upgrade Factfinding Module

- Designed a web application that combines LLM AI's comprehensive abilities to account for the user's personal interests and processed financial data to guide goal-setting process.
- Navigated housing laws and policies to create accurate and realistic data projections.
- Bi-weekly design sprints with a SCRUM agile workflow.
 - Tech stack: Google Cloud Run, Postgre SQL, Docker, React, FastAPI, OpenAI API

SUTDHotelBooking - Hotel Booking Site Case Study

- Delivered functioning site with user authentication, login system, API fetching, and full hotel searching capabilities with fuzzy searching.
- Led team of 6 through the process of **client interaction**, milestone setting, and determining project parameters for a frontend-focused full-stack web application development.
- Used Figma for **rapid prototyping** and mocking of the site.
- Implemented **testing** structures (unit/integration/E2E) using **Jest** and **Cypress**.
 - Tech stack: NextJS, React, TailwindCSS, NextAuth, Prisma, Typescript, and MySQL.

FPGA Arcade Game and ALU design

- Designed a working **16-bit ALU** and **CPU** following MIT's Beta CPU ISA using Lucid HDL, deployed on the Xilinx Artix-7 **FPGA**.
- Wrote **graphics driver** for our custom CPU design to display graphics on an RGB LED matrix.
- Prototyped **game code in C** and manually transpiled C code to the Beta assembly language to run on our CPU implementation on the FPGA.

ResiCo - Mobile app to digitalize residential notice boards

- Won third place in the Singtel-Information Systems and Programming Merit Award.
- Led team of 7 through a software design and analysis process for a mobile application with a focus on good software collaboration practices.
- Led the developer team with frontend development using **Android Studio** and **Java**, with strict adherence to using Google's new **Material 3 components**.
- Backend development using **Firebase**, rapid iterative prototyping using **Figma**.

Skills

Full Professional Proficiency

- Java
- Python
- JavaScript
- HTML/CSS
- ReactJS
- C#
- C/C++
- Git
- TypeScript
- Figma

Decent Proficiency

- Lua
- Flask (Python)
- MySQL
- Docker

In Progress

- AWS
- Azure
- MongoDB
- Google Cloud

Soft Skills

- Presentations/Teaching
- Team leadership
- Adaptable
- Resourceful
- Proficient in Mandarin and English

Related Coursework

- Computational Structures
- Computer System Engineering
- Algorithms
- Information Systems
- Elements of Software Construction
- Machine Learning
- Applied Deep Learning
- Extended Reality
- Graphics and Visualisation
- Programming Language Concepts

Education

Bachelor of Engineering in Computer Science and Design

Singapore University of Technology and Design

2021-2025 • CGPA 4.50

Executive member of Cycling IG

- Organized and hosted weekly biking activities
- Managed club finances

Member of the SUTD XRCommunity

- Organized club meetings and presented topics on current events in XR (Extended Reality)

Hobbies & Interests

- Portrait and Travel Photography
- 3D Modelling with Blender
- Road cycling
- VR gaming and game development
- Rollercoaster rides and whitewater rafting
- Avid manga reader